

Hong-Sheng Lai

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Education

Johns Hopkins University

PhD in Computer Science (*advised by Dr. Steven Salzberg & Dr. Mihaela Pertea*)

Baltimore, MD

Aug 2026 – Present

Carnegie Mellon University

MS in Computational Biology (*advised by Dr. Carl Kingsford*)

Pittsburgh, PA

Aug 2024 – Dec 2025

National Taiwan University

BS in Biomechatronics Engineering, Minor in Economics (*advised by Dr. Chien-Yu Chen*)

Taipei, Taiwan

Sep 2019 – Jun 2023

Skills

- **Programming Languages:** Python, C++, Java, SQL, R, LaTeX
- **MLOps & Framework:** PyTorch, JAX, Grafana, Wandb
- **Developer Tools:** Git, Linux, Docker, Slurm, Kubernetes, CI/CD, MySQL, MongoDB, Apache Airflow

Work Experience

Genentech

South San Francisco, CA

Computational Biology Intern

Jan 2026 – Jul 2026

- Implemented a local **agentic AI** assistant using **LangGraph (ReAct)**, **LangChain**, and **Ollama** to let scientists configure sequencing analysis pipelines via natural language, featuring structured tool use with **Pydantic** schemas, real-time GUI control via a socket bridge, and on-premises inference for data privacy.
- Designed **Claude Code** skills for automating scientific document generation across 10 life-science domains, using structured prompt pipelines with multi-step citation verification (Crossref API), template-driven output, and built-in compliance checks.
- Developed **NGS** quality control pipelines for **cell therapy** materials in **long-read sequencing**, automating consensus generation, variant interpretation, and impurity inference, with results rendered in HTML/PDF templates.

Taiwan AI Labs

Taipei, Taiwan

Machine Learning Engineer Intern

Jul 2023 – Jan 2024

- Enabled sequence-function similarity search in **Qdrant vector database** with **contrastive learning** and **genomic language models**.
- Accelerated sparse matrix regression models with **JAX** for **federated analysis**.
- Established an end-to-end clinical annotation pipeline for **pharmacogenomics**-based drug usage recommendations.

Missmi Intelligence

Taipei, Taiwan

Software Engineer

Jun 2021 – Jun 2023

- Built and launched an NGS gene reporting product, collaborating with medical experts on action plan suggestions; over 400 units sold.
- Automated PDF report generation by integrating secondary analysis results into HTML templates, orchestrated with **Apache Airflow**.

Projects

ARCADE: Controllable Codon Design from Foundation Models via Activation Engineering

ISMB – MLCBS, 2026. [\[Paper\]](#)

- Engineered a controllable codon design model by applying **activation engineering** techniques to both **decoder-only** and **encoder-decoder language models** to achieve **multi-objective optimization**.

Alignment-Based Copy Number Estimation and Allele Typing in Complex Genes

In submission. [\[TypeAssembly\]](#) [\[deCYPher\]](#)

- Developed **local alignment** algorithms for **copy number estimation** and **star-allele typing** in structurally complex gene regions (pharmacogenes, KIR, HLA) from haplotype-resolved **long-read assemblies**.